

Single gold nanorods as quantum-plasmonic antennas from the visible to the telecom bands: local density of states, hot spots and pulsed single-photon emission

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Abstract

Plasmonic nanoantennas can accelerate the emission of single quantum emitters by three orders of magnitude, but whether the emitted light keeps its quantum properties depends on the coupling regime, on the excitation scheme and on the wavelength. We map the local density of optical states (LDOS) and the field hot spots around single gold nanorods with converged boundary-element calculations, tune the longitudinal resonance from 760 nm to the telecom O and C bands (1310–1550 nm) in a glass-like host, and evaluate the quantum figures of merit of a resonance-matched emitter with a master-equation model of pulsed excitation. The radiative LDOS is concentrated within about 10 nm of the tips, where an axial dipole radiates more than two orders of magnitude faster than a transverse one, while the antenna efficiency is set by the rod diameter rather than by the wavelength: 22–25 % for 20 nm, 44–50 % for 30 nm and about 60 % for 40 nm rods across 760–1660 nm. A rod resonating at 1550 nm raises the radiative rate of an emitter by 10^3 at a 5 nm gap, yet emitter and plasmon remain weakly coupled ($g \leq 0.43 \kappa/4$). The Purcell-shortened lifetime, 0.2–3 ps, sets the quantum performance: with 100 fs excitation pulses re-excitation limits the single-photon purity to $g^{(2)}(0) = 0.15$ at a 5 nm gap and 0.02 at 20 nm, while at cryogenic dephasing the indistinguishability reaches 0.88–0.98; at room temperature it stays below 0.15. The results give quantitative design rules for plasmonic single-photon sources at telecom wavelengths.

Keywords: quantum plasmonics; local density of optical states; gold nanorod; telecom single-photon source; Purcell effect; indistinguishability; boundary element method.

1 Introduction

Quantum communication requires sources of single, indistinguishable photons at the wavelengths of the fibre network, the O band near 1310 nm and the C band near 1550 nm [1, 2]. The spontaneous emission rate of an emitter is proportional to the local density of optical states (LDOS) at its position [3, 4], and metal nanoparticles, whose localized surface plasmons confine light far below the diffraction limit, raise the LDOS by orders of magnitude [5–7]. A large rate enhancement makes an emitter brighter and, because the radiative lifetime becomes short compared with the dephasing time, can make its photons indistinguishable [8]. The price is absorption in the metal, which turns part of the enhanced decay into heat [9, 10], and the photons are useful only if the emitter remains a two-level system: in the strong-coupling regime [11, 12] or under re-excitation within one excitation pulse [13] the statistics of the emitted light change.

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Most theoretical studies of plasmonic single-photon sources have considered visible wavelengths and coupled structures, such as nanorod dimers treated with quantized quasinormal modes [14–16], while a telecom-band source has recently been demonstrated with colloidal quantum dots in a nanoparticle-on-mirror cavity [17]. Single gold nanorods are simpler to synthesise, their longitudinal resonance can be tuned from the visible to the infrared through the aspect ratio, and the field at their tips is enhanced by the lightning-rod effect [18, 19]. Here we ask four questions for single gold nanorods: (i) where around the rod the LDOS and the field are concentrated, and how this depends on the orientation of the emitter; (ii) how the enhancement and the antenna efficiency change when the rod is tuned from the visible to the telecom bands; (iii) whether a single emitter remains weakly coupled; and (iv) how the Purcell-shortened lifetime translates into single-photon purity and indistinguishability under realistic pulsed excitation. The electromagnetic calculations use the boundary-element method validated in a companion paper [20] against the exact Mie solution to 1 %, against two finite-difference time-domain solvers and against single-molecule experiments.

2 System and methods

Antenna and emitter. Gold nanorods with hemispherical caps (diameter D , total length L) are embedded in a homogeneous medium of index $n = 1.45$, representing a polymer or glass matrix in which the emitter is embedded, or in air for the LDOS maps. The emitter is a point dipole at distance d (the gap) from the surface; its orientation is axial ($\parallel$) or transverse ($\perp$) to the rod axis unless stated otherwise. Gold is described by the Johnson–Christy permittivity [21].

Electromagnetics. The total decay rate relative to that in the homogeneous medium is the Purcell factor,

$$F_p = \frac{\Gamma}{\Gamma_0} = \frac{6\pi c}{\omega n^2} \hat{\mathbf{n}} \cdot \text{Im} \overleftrightarrow{\mathbf{G}}(\mathbf{r}_0, \mathbf{r}_0; \omega) \cdot \hat{\mathbf{n}} = \frac{\rho_{\hat{\mathbf{n}}}(\mathbf{r}_0, \omega)}{\rho_0(\omega)}, \quad (1)$$

i.e. the partial LDOS at the emitter normalised to its value in the host [4]; the radiative part T (the radiative LDOS) is the power reaching the far field, and $\eta_a = T/F_p$ is the antenna efficiency. Both were computed with the retarded boundary-element method with curved elements (MNPBEM [22], element size 2–4 nm); the convergence and the validation of the identical set-up are documented in Ref. [20].

Coupling regime. For a single dominant plasmon mode of energy width $\hbar\kappa$, the coupling to an emitter with transition dipole $\boldsymbol{\mu}$ is $\hbar g = \boldsymbol{\mu} \cdot \mathbf{E}_{\text{vac}}(\mathbf{r}_0)$, proportional to the vacuum field of the mode at the emitter; in the bad-cavity limit the mode-induced decay rate is $4g^2/\kappa = F_p\Gamma_0$, hence $\hbar g = \frac{1}{2}\sqrt{F_p\hbar\Gamma_0\hbar\kappa}$, and strong coupling requires $g > \kappa/4$ [11]. The LDOS maps are therefore also maps of g^2 .

Pulsed emission. The emitter is a two-level system with total decay rate $\Gamma = F_p\Gamma_0$ (intrinsic quantum yield one), pure dephasing rate γ^* and a resonant Gaussian π pulse of intensity full width τ_p . We solve the Lindblad master equation with QuTiP [23] and obtain, from the two-time correlation functions $G^{(1)}(t, \tau) = \langle \sigma_+(t+\tau)\sigma_-(t) \rangle$ and $G^{(2)}(t, \tau) = \langle \sigma_+(t)\sigma_+(t+\tau)\sigma_-(t+\tau)\sigma_-(t) \rangle$, the photon number per pulse N , the single-photon purity $g^{(2)}(0) = 2 \iint G^{(2)} dt d\tau / N^2$ and the two-photon interference visibility $I = 2 \iint |G^{(1)}|^2 dt d\tau / N^2$ [13]. In units of $1/\Gamma$ the results depend only on $\tau_p\Gamma$ and γ^*/Γ .

3 Results

3.1 LDOS and hot spots

Figure 1 maps the LDOS, the radiative LDOS, the antenna efficiency and the local intensity around a 60 nm × 20 nm rod in air at its longitudinal resonance (608 nm). The total LDOS is large everywhere within a few nanometres of the metal: this is the non-radiative near field of the

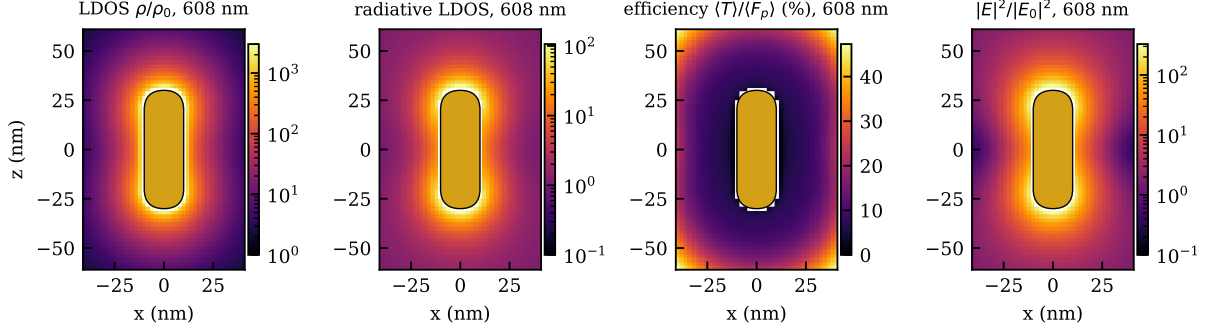


Figure 1: Gold nanorod 60 nm×20 nm in air at its longitudinal resonance (608 nm), xz plane (BEM): orientation-averaged LDOS $\rho/\rho_0 = \langle F_p \rangle$, radiative LDOS $\langle T \rangle$, antenna efficiency $\langle T \rangle / \langle F_p \rangle$ and intensity enhancement for a plane wave polarised along the rod.

lossy surface, which quenches the emitter. The radiative LDOS and the intensity, in contrast, are concentrated at the tips: 4 nm beyond the apex an axial dipole has $F_p = 2.2 \times 10^3$ and $T = 157$ and the intensity is enhanced 158-fold, whereas 4 nm from the side of the rod $T \leq 33$ and the intensity enhancement is 32. The volume in which the intensity is enhanced more than 100-fold is about $5 \times 10^3 \text{ nm}^3$, a third of the rod volume. The efficiency map shows the counterpart: for an orientation-averaged emitter the efficiency is below 5 % within 2 nm (side) to 4 nm (tip) of the surface and rises only slowly with the distance, to 8 and 12 % at 10 and 20 nm beyond the tip, because the orientations that do not couple to the longitudinal mode are quenched. The LDOS is strongly anisotropic. At the apex the transverse dipole has $F_p = 69$ and $T = 0.26$, so that the axial-to-transverse ratio is 32 for F_p and 600 for T ; at the side of the rod the orientation along the axis is again favoured (radiative LDOS 33 versus 2.9 for the normal orientation), because the longitudinal mode produces a field parallel to the surface there. Because $g^2 \propto F_p$, the coupling to the longitudinal mode is largest at the tips and for dipoles aligned with the local mode field.

3.2 From the visible to the telecom bands

In a host of index 1.45, rods with $D = 20 \text{ nm}$ resonate (radiative peak of the axial dipole) at 760 nm ($L = 60 \text{ nm}$), 830 nm (70 nm), 1280 nm (140 nm) and 1510 nm (180 nm) (Fig. 2a). The enhancement grows with the wavelength: at a 5 nm gap the radiative enhancement rises from 464 at 760 nm to 1.2×10^3 at 1510 nm and F_p from 2.2×10^3 to 5.2×10^3 ; the axial-to-transverse ratio at the resonance is 230–330 at 5 nm and 450–850 at 10 nm (Fig. 2b). The antenna efficiency, however, stays between 22 and 25 % for all rods (Fig. 2c). Two effects compensate: the absorption of gold decreases towards the infrared, but at fixed diameter the rod volume shrinks relative to λ^3 , and with it the radiative damping of the mode. The efficiency is therefore controlled by the diameter: at the same resonances, rods of 30 nm diameter reach 44–50 % and rods of 40 nm 59–66 %, at the cost of a smaller F_p (2.4×10^3 for $D = 30 \text{ nm}$ and 1.4×10^3 for 40 nm at 5 nm and about $1.5 \mu\text{m}$) and a smaller anisotropy (110–120 and 53–58). For an emitter in the C band a rod of $D = 30 \text{ nm}$ and $L = 240 \text{ nm}$ (radiative peak 1520 nm) gives $T = 1.1 \times 10^3$ at 5 nm and 450 at 10 nm with $\eta_a \approx 46 \%$.

3.3 Coupling regime

Taking $\hbar\kappa$ from the width of the radiative peak and $\Gamma_0^{-1} = 1 \text{ ns}$, an upper bound on the bare rate of molecular and solid-state emitters, the coupling of a resonance-matched emitter at a 5 nm gap is $\hbar g = 5.2 \text{ meV}$ at 810 nm, 7.5 meV at 1310 nm and 8.1 meV at 1550 nm, i.e. $g/(\kappa/4) = 0.29, 0.36$ and 0.43 (Table 1). The coupling increases towards the infrared because the plasmon linewidth decreases, but all configurations remain in the weak-coupling regime, as do slow emitters such

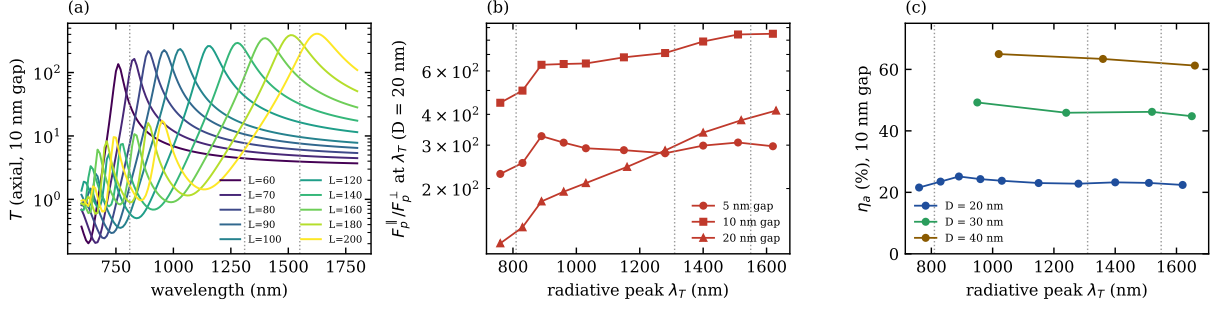


Figure 2: Gold nanorods in a host of index 1.45 (BEM, axial dipole on the axis). (a) Radiative enhancement at a 10 nm gap for $D = 20$ nm and $L = 60$ – 200 nm; dotted lines: 810, 1310 and 1550 nm. (b) Axial-to-transverse ratio of F_p at the radiative peak. (c) Antenna efficiency at the radiative peak for $D = 20, 30$ and 40 nm.

Table 1: Resonance-matched gold nanorods ($D = 20$ nm, $n = 1.45$, axial dipole) for a bright emitter ($q_0 = 1$, $\Gamma_0^{-1} = 1$ ns). $g^{(2)}(0)$ and I for 100 fs pulses and $\hbar\gamma^* = 1$ μ eV (4 K); in brackets: 10 meV (room temperature).

λ_e (nm)	gap (nm)	T	F_p	η_a	Γ^{-1} (ps)	$g/(\kappa/4)$	$g^{(2)}(0)$	I
810	5	518	2305	22%	0.43	0.29	0.08	0.93 (0.07)
810	20	30	125	24%	8.0	0.06	0.005	0.97 (0.004)
1310	5	942	4160	23%	0.24	0.36	0.13	0.89 (0.11)
1550	5	1201	5305	23%	0.19	0.43	0.15	0.88 (0.13)
1550	10	396	1736	23%	0.58	0.26	0.07	0.94 (0.05)
1550	20	89	378	24%	2.6	0.12	0.02	0.98 (0.01)

as colloidal PbS quantum dots ($\Gamma_0^{-1} \sim 1$ μ s, $g/(\kappa/4) \leq 0.014$). The rate description and the two-level model below are therefore justified; strong coupling of single emitters would require sub-nanometre gaps [12].

3.4 Pulsed single-photon emission

At a 5 nm gap the Purcell factor shortens a 1 ns radiative lifetime to 0.43 ps at 810 nm and 0.19 ps at 1550 nm; at 20 nm to 8 and 2.6 ps. Such lifetimes are comparable to the duration of available excitation pulses, so that the emitter can decay and be re-excited within one pulse. Figure 3a shows the resulting purity: $g^{(2)}(0)$ grows almost linearly with $\tau_p\Gamma$, from 0.009 at $\tau_p\Gamma = 0.02$ to 0.15 at 0.5 and 0.33 at 2, independently of the dephasing. With 100 fs pulses the rod at 1550 nm gives $g^{(2)}(0) = 0.15$ at a 5 nm gap, 0.065 at 10 nm and 0.016 at 20 nm; with 1 ps pulses the values are 0.52, 0.30 and 0.12 (Table 1). The visibility of two-photon interference decreases with the pulse length even without dephasing (Fig. 3b), from 0.99 at $\tau_p\Gamma = 0.02$ to 0.86 at $\tau_p\Gamma \geq 1$, because re-excitation produces photons with random emission times. Pure dephasing reduces I further, close to the bad-cavity result $\Gamma/(\Gamma + 2\gamma^*)$ [8]. For $\hbar\gamma^* = 1$ μ eV, typical of quantum dots at 4 K, the 1550 nm antenna gives $I = 0.88$ (5 nm gap), 0.94 (10 nm) and 0.98 (20 nm) with 100 fs pulses; at room temperature ($\hbar\gamma^* \approx 10$ meV) I stays at 0.01–0.13 even for the largest Purcell factor. These results agree with the conclusion of quantized-quasinormal-mode calculations for nanorod dimers that sub-picosecond pulses are required [16], and quantify it for single rods: the gap that maximises the rate does not maximise the purity, and a gap of 10–20 nm, combined with ~ 100 fs pulses, gives $g^{(2)}(0) \leq 0.07$ and $I \geq 0.94$ at cryogenic temperature while retaining a radiative enhancement of 90–400.

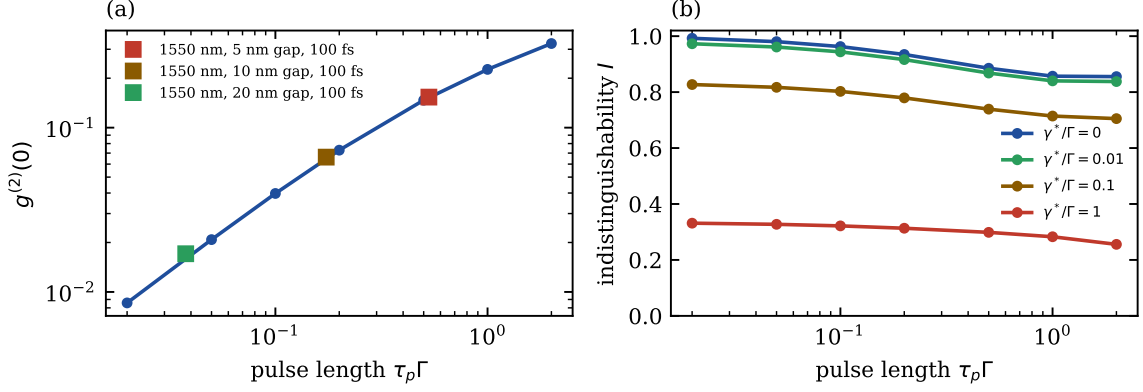


Figure 3: Pulsed excitation of a Purcell-enhanced two-level emitter (master equation). (a) $g^{(2)}(0)$ versus pulse length in units of the lifetime; squares: resonance-matched rods at 1550 nm with 100 fs pulses. (b) Two-photon interference visibility versus pulse length for different dephasing rates.

4 Discussion

The calculations separate three quantities that are often conflated in the term “Purcell enhancement”. The total LDOS sets the speed of the emitter and is largest in a thin, non-radiative shell around the whole particle; the radiative LDOS sets the brightness and is concentrated at the tips; and the coupling strength g , proportional to the mode field, sets the coupling regime, which remains weak for single emitters near single rods at all wavelengths considered. For telecom single-photon sources the main design parameters are the rod diameter, which fixes the antenna efficiency, the length, which fixes the wavelength, and the gap, which trades the radiative enhancement against the purity under pulsed excitation. The orientation of the transition dipole matters as much as its position: a transverse dipole at the tip is quenched rather than enhanced. At cryogenic temperature the antenna can make the photons of a dephased emitter nearly indistinguishable if the excitation pulses are much shorter than the Purcell-shortened lifetime; at room temperature the dephasing of solid-state emitters is too fast even for Purcell factors of 10^3 – 10^4 , in agreement with Ref. [8].

The model neglects the background emission of gold, which competes with the single-photon signal for weakly absorbing emitters [20, 24]; the semiconductor environment of epitaxial telecom quantum dots, whose high index would require a rod on the semiconductor surface rather than a homogeneous host; and phonon sidebands, which reduce the indistinguishability of quantum dots further. A quantized quasinormal-mode treatment [15, 16] would include the frequency dependence of the plasmon response within the pulse bandwidth, which is small here because the plasmon linewidth (70–85 meV) exceeds the bandwidth of a 100 fs pulse (18 meV).

5 Conclusion

Single gold nanorods tuned from the visible to the telecom O and C bands raise the radiative rate of a resonance-matched emitter by up to 10^3 while keeping it weakly coupled. Their antenna efficiency is set by the diameter (23 % for 20 nm, 46 % for 30 nm, about 60 % for 40 nm) rather than by the wavelength, and the radiative hot spots are confined to about 10 nm around the tips, where only dipoles aligned with the mode field are enhanced. Because the Purcell-shortened lifetime reaches 0.2–3 ps, sub-picosecond excitation pulses and gaps of 10–20 nm are needed for a single-photon purity better than $g^{(2)}(0) \approx 0.07$, and indistinguishability above 0.9 is reached only at cryogenic dephasing rates.

Data availability

Simulation scripts, raw spectra and analysis code are available at <https://github.com/aminsadidi/maghale>.

Author contributions

[to be completed by the authors.]

Conflict of interest

The authors declare no competing interests.

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